

BHAVJEET SINGH

github.com/bhavjeetsingh | linkedin.com/in/bhavjeet-singh | bhavjeetsingh784@gmail.com | +91 7678205626 | Delhi

PROFESSIONAL SUMMARY

AI/ML Backend Engineer from IIT Madras (B.Sc. Data Science) with production experience building LLM applications, RAG pipelines, and high-impact ML systems deployed on AWS and Azure. Proficient in **Python, PyTorch, SQL, and Docker**. Fine-tuned a local 3B model to outperform GPT-4o-Mini, reduced legal contract review time by 60%, and optimized demand forecasting to save INR 62 crore/hour.

EDUCATION

Indian Institute of Technology, Madras <i>Bachelor of Science in Data Science and Applications</i>	2025
--	------

PROJECTS

- ToolCaller-India – Local Function-Calling LLM** | *Python, PyTorch, Transformers, LoRA/PEFT, Hugging Face, Gradio* 
- Engineered and fine-tuned a **3B parameter local LLM (ToolCaller-India)** using **PyTorch** and **LoRA** to execute complex, low-latency function calls on Indian-context APIs (UPI, PNR, GSTIN, Aadhaar, BBPS).
 - Synthesized and compiled a strict **13.3k SFT training mixture (8.8k custom multi-turn Indian examples)** using data deduplication and schema validation to enforce exact tool structures.
 - Achieved **70.39% weighted accuracy** on the BFCL dev split (outperforming GPT-4o-mini's **69.07%**) with **100% JSON formatting compliance** and 0% syntax errors, deploying a responsive Gradio dashboard.
- ContractIQ – Legal AI Data Pipeline Service** | *Python, FastAPI, LangChain, FAISS, Docker, AWS ECS/Fargate* 
- Architected a scalable **RAG pipeline** using FastAPI, LangChain, and FAISS to parse and index **500+ unstructured legal PDFs** for term extraction.
 - Optimized **LLM inference by 40%** using vLLM quantization and Redis caching, integrating **LangSmith** for token and latency observability.
 - Cut average legal contract review time by **60%** while maintaining a P50 latency of **2.79s** via Gemini routing, achieving an efficient operational cost of **\$0.08 per 67k tokens**.
- 24-Hour Electricity Demand Prediction – ML Pipeline** | *Python, PyTorch, XGBoost, Prophet, LSTM, MLflow, FastAPI* 
- Forecasted **India's 180 GW electricity demand** 24 hours ahead, achieving **6.91% MAPE** – saving **INR 62 crore/hour** in grid settlement penalties.
 - Engineered **19k+ hourly features** (lag, weather, stats) using **PySpark** to optimize transformations across historical grid datasets.
 - Tracked **14+ MLflow runs** with full reproducibility and **deployed automated CI/CD pipelines** using GitHub Actions to Azure.

TECHNICAL SKILLS

Languages: Python, PyTorch, SQL, Bash, Git/GitHub
GenAI & LLM: LoRA Fine-Tuning, PEFT, Hugging Face (Transformers), LangGraph, LangChain, CrewAI, RAG, AI Agents, Tool/Function Calling, Prompt Engineering, vLLM
LLM APIs & Observability: OpenAI, Anthropic, Gemini, Groq, LangSmith, FAISS, Qdrant, Pinecone, DataStax Astra DB
Backend & APIs: FastAPI, REST APIs, Async/Await, Pydantic, CRUD Operations, MCP
Cloud & DevOps: AWS (ECS, Fargate, EC2), Azure, Docker, GitHub Actions, CI/CD
Databases: PostgreSQL, Redis, PySpark, pandas, NumPy
Frontend: React.js, HTML/CSS, Tailwind CSS

CERTIFICATIONS

LLMOps Live Cohort Program	December 2025 – April 2026
Ultimate Live RAG Bootcamp: Agentic Systems with Cloud Deployment	Jan 2026 – April 2026